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SUPERSEDING
MIL-C-29109A
23 May 1978

MILITARY SPECIFICATION

COVERALLS, ANTI-EXPOSURE

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a hooded submarine deck exposure coverall constructed with a coated nylon fabric outershell, a plastic foam interlining, and a nylon fabric lining.

1.2 Classification. The coverall shall be of one type only, and in the following sizes (see 6.2):

Schedule of sizes

| | |
|---|---|
| Small (34-36) | Large (42-44) |
| Medium (38-40) | X-Large (46-48) |
| XX-Large (50-52) | |

2. APPLICABLE DOCUMENTS

2.1 Government documents.

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| V-F-106 | - Fasteners, Slide, Interlocking |
| V-T-295 | - Thread, Nylon |
| JJ-W-155 | - Webbing, Textile, (Cotton, Elastic) |
| * NN-P-71 | - Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial) |
| DDD-L-20 | - Label; For CLothing, Equipage, and Tentage (General Use) |
| DDD-B-1199 | - Binding, Textile, Cotton, Bias-Cut |

MILITARY

| | |
|---|---|
| MIL-P-12420 | - Plastic Material Cellular, Elastomeric |
| * MIL-P-15011 | - Pallet, Material Handling, Wood Post Construction, 4-Way Entry |
| MIL-B-17757 | - Boxes, Shipping, Fiberboard (Modular Sizes) |
| MIL-C-19759 | - Cloth, Coated, Twill, Nylon (Low Count) |
| MIL-F-21840 | - Fastener Tapes, Hook and Loop, Synthetic |
| MIL-C-21852 | - Cloth, Nylon, Taffeta (2.0 ounce) |
| MIL-C-43256 | - Cord, Polyester, Solid Braid |
| MIL-C-43836 | - Cloth, Interlining, Non-Woven |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-191 | - Textile Test Methods |
| FED-STD-751 | - Stitches, Seams, and Stitchings |

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-147 | - Palletized Unit Loads |
| MIL-STD-1668 | - Provisions for Evaluating Quality of Cloth Coveralls |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D1056-78 Flexible Cellular Materials-Sponge or Expanded Rubber

(Copies of ASTM Publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606-5808).

THE COLOR ASSOCIATION OF THE UNITED STATES

Department of Defense Standard Shades for Buttons
Department of Defense Standard Shades for Sewing Threads

(Application for copies of color cards should be addressed to the Color Association of the U.S., 343 Lexington Avenue, New York, NY 10016).



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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Basic material. The basic material for the outer shell shall be chloroprene coated nylon conforming to Type I of MIL-C-19759, except that the uncoated side of the finished (coated) cloth shall match Shade Orange 3604, piece number 75345. The coating compound shall be pigmented orange. In addition, tear strength in the filling direction shall be a minimum of 3400 grams, hydrostatic resistance shall be after 40 hours exposure in the weatherometer and the flex stiffness shall be tested at -20°F.

* 3.3.2 Lining. The material for the coverall lining shall be nylon taffeta conforming to type III of MIL-C-21852 except that the color may be olive green, black or approximately the same shade as the basic material.

3.3.3 Interlining. The material for the interlining shall be plastic foam with a thickness of 1/8 inch and 1/4 inch conforming to type II, class 6 of MIL-P-12420, except that the buoyancy shall not be less than 54.0 pounds per cubic foot when tested as specified in 4.4.1.

* 3.3.4 Thread. The thread for seaming and stitching the coverall shall be nylon, conforming to type I, class B, letter sizes B and AA of V-T-295. The color shall be orange TH, Cable No. 66074 in accordance with DOD Standard Shades for Sewing Threads. For overall lining, the color of the thread shall approximate the shade of the Taffeta used.

3.3.4.1 Colorfastness. The dyed thread shall show colorfastness to weathering and laundering equal to or better than the standard sample when tested as specified in V-T-295. As a limit of acceptability, or when no standard sample is available, the dyed thread shall show a minimum of "good" fastness to weathering and laundering when tested as specified in V-T-295.

3.3.5 Slide fasteners.

* 3.3.5.1 Utility pocket closure. THe slide fastener shall be 5 inches molded plastic individual element or continuous element monofilament conforming to type I, style 3, size MS, short tab pull of V-F-106. The color of the plastic slide fastener components shall be black.



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3.3.5.2 Front closure. The slide fastener shall be molded plastic individual element or continuous element monofilament conforming to type I, style 15, size MHS, long tab pull of V-F-106. The color of the plastic slide fastener components shall be black. The length of the slide fastener shall be such that the top stop of the slide fastener shall be set not more than 1/4 inch from the finished neck seam with the bottom stop of the slide fastener set as indicated on patterns.

* 3.3.5.3 Sleeve closure. The slide fastener shall be 7 inches long, molded plastic individual element or continuous chain, conforming to type I, style 3, size MS, short tab pull of V-F-106. The color of the plastic slide fastener components shall be black.

* 3.3.5.4 Tape. The tape for all the slide fasteners shall be cotton, synthetic or a combination of both. The color of the tape shall be black or orange and shall show good colorfastness to laundering (see 4.4.1).

* 3.3.5.5 Thong. The thong for the front closure and utility pocket shall be a nylon braid or webbing approximately 3/8 inch in width with a finished hanging length of 4-1/2 to 5-1/2 inches for front closure and 1-1/2 to 2 inches for utility pocket. The color of the thong shall be olive green, black, or orange and shall show good colorfastness to laundering (see 4.4.1).

* 3.3.6 Fastener tapes. The matching hook and loop fastener tapes shall be 1 inch wide for the waist adjustment straps and neck seal attachment, and 2 inches wide for the sleeve, neck seal, neck tab, and leg adjustments, black or orange in color, conforming to type II, class 3 of MIL-F-21840.

* 3.3.7 Cord. The polyester cord for the hood adjustment drawcord shall be 3/16 inch diameter, natural, olive green or black, conforming to MIL-C-43256, except that when dyed it shall also show good colorfastness to crocking (see 4.4.1). Ends of the cut cord shall be treated chemically with a clear resin finish to prevent unravelling.

3.3.8 Identification and instruction label. Each coverall shall have a combination identification, size, and instruction label conforming to type I, classes 1, 2, and 3 of DDD-L-20. The printing shall show fastness to laundering. The labels shall bear the following inscription:



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COVERALLS, ANTI-EXPOSURE
CONTRACT NO: DLA-100-00-0-0000 (Example)
STOCK NO: 8415-00-000-0000 (Example)
SIZE: Medium 38-40 (Example)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (If other than contractor):

Hand Launder With Warm Water and Mild Soap
Hang Dry Inside Out

OR

Follow NAVEDTRA Manual 414-01-45-81 Formula III
Tumble Dry Low Heat

NOTE:

TURN COVERALL INSIDE OUT BEFORE LAUNDERING
HIGH LAUNDERING AND DRYING TEMPERATURES WILL DESTROY FOAM INTERLINING MATERIAL
ALL HOOK AND LOOP CLOSURES MUST BE SECURELY FASTENED OR THEY WILL DESTROY LINING MATERIAL

3.3.9 Elastic webbing. The elastic webbing for the waistband adjustment strap shall be cotton conforming to type I, class 3, 3/4 inch wide of JJ-W-155.

* 3.3.10 Adhesive cement. The cement used for adhering butted seams of the foam interlining and neck seal and taping of butted joints of the foam interlining shall be a chloroprene compound. The use of reclaimed rubber is prohibited. The butted seams shall have a seam strength of 10 pounds per 2 inch width when tested as specified in 4.4.1.2.

* 3.3.10.1 Cemented seams. All butted seams of the foam interlining shall have a minimum breaking strength of 10 pounds per 2 inch width when tested as specified in 4.4.1.5.2.

* 3.3.10.2 Tape (cover). The tape used to cover all butted foam seams shall be 3/4 inch bias cut strips of the basic material (see 3.3.1).

3.3.11 Tape. The cotton material used for extending the foam interlining shall conform to type II, class 1 of DDD-B-1199.

3.3.12 Pocket welt interlining. The pocket welt interlining shall be a non-woven fabric conforming to type II, class 1 of MIL-C-43836.

* 3.3.13 Seal.

* 3.3.13.1 Chloroprene foam. The choloprene foam for the neck seal shall be pigmented black, gas expanded chlorinated cellular rubber, having a closed cell structure conforming to ASTM Classification RE-41-E1 Test Method D-1056-78. The foam material shall be 1/8 (± 1/16) inch thick with skin on one side only (see 4.4.1).



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* 3.3.13.2 Knitted stretch cloth. The stretch cloth shall be circular knit from 100% continuous filament nylon yarn and shall conform to the physical requirements below (see 4.4.1).

Knitted Stretch Cloth

| Knit | Yarns per inch (min.) Wales | Courses | Weight oz/yd$^2$ | Percent elongation Wales | Courses | Color |
|---|---|---|---|---|---|---|
| Jersey | 36 | 38 | 2.4 ± 0.2 | 150 min<br>225 max | 225 min<br>400 max | Black |

* 3.3.13.3 Colorfastness. The knitted cloth shall show "good" fastness to laundering and perspiration equal to or better than the standard sample. As a limit of acceptability, when no standard sample is available the finished material shall show a minimum of "good" fastness to laundering, perspiration, and a Munsell value for dry and wet crocking not lower than 8.5 (see 4.4.1).

* 3.3.13.4 Laminated foam. The knitted stretch cloth shall be laminated to the cut surface side of the chloroprene foam by means of a suitable adhesive, or other process capable of meeting the physical requirements below. The adhesive used shall be non-toxic, non-irritant and free from objectionable odors. The fabric backed (laminated) side shall be smooth and free of wrinkles or puckers and show no signs of blocking, tackiness or adhesive strike-thru and shall conform to the requirements below, when tested as specified in 4.4.1.

Laminated foam

| Characteristic | Requirement |
|---|---|
| Laundering resistance | No delamination |
| Adhesion, initial (lb/2 in) (min) | 8.0 |
| Adhesion, after laundering (lb/2 in) (min) | 8.0 |
| Adhesion, after aging (lb/2 in) (min) | 8.0 |
| Blocking | No blocking |

* 3.3.14 Bias tape. The bias tape used on the neck seal shall be a good commercial grade, colored black, and 1-7/8 inches wide.

* 3.4 Design. The impermeable coveralls shall have an outershell of coated nylon fabric with the coating finishing on the underside. The coverall shall be lined with an uncoated nylon taffeta and interlined with a layer of expanded plastic foam. The upper front and back sections shall be insulated with 1/4 inch thick foam. Remainder of coverall shall have 1/8 inch thick foam. The coverall shall have an attached hood with drawcord. The sleeve and leg bottoms and waistband shall have hook and loop tape fastener adjustments. Coverall shall have a welted open pocket at each hip and a zippered utility pocket on the left sleeve. Front closure shall have a slide fastener from neck to crotch with a protective flap. The coveralls shall have a chloroprene foam neck seal, a slide fastener sleeve adjustment and a neck tab.



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3.4.1 Figure. Figure 1 is furnished for information purposes only. When inconsistencies exist between the written specification and the figure, the written specification shall govern.

* 3.5 Patterns. Standard patterns used to cut working patterns will be furnished by the Government (see 6.3). The patterns allow 3/8 to 1/2 inch for all seams, except as otherwise indicated in Table I. The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way, except that additional notches for use during construction are allowed on the working patterns.

* 3.5.1 Pattern parts. The component parts shall be cut from the material specified in accordance with the following pattern parts:

| Material | Pattern parts | Cut parts |
|---|---|---|
| Cloth, coated, Twill, nylon (outershell) | Back upper | 1 |
| | Back trouser | 2 |
| | Front | 2 |
| | Top sleeve | 2 |
| | Undersleeve | 2 |
| | Front facing | 2 |
| | Back waist channel | 1 |
| | Pocket (inner) | 2 |
| | Pocket (outer) | 2 |
| | Long utility pocket (left sleeve) | 1 |
| | Short utility pocket (left sleeve) | 1 |
| | Narrow accessory pocket (left sleeve) | 1 |
| | Wide accessory pocket (left sleeve) | 1 |
| | Pocket welt (bottom ply) | 2 |
| | Pocket welt (top ply) | 2 |
| | Adjustment tab (sleeve bottom) | 2 1/ |
| | Adjustment tab (leg bottom) | 2 1/ |
| | Adjustment tabs (waistband) | 1 1/ |
| | Hood, side panel | 2 |
| | Hood, center panel | 1 |
| | Hood, cord channel | 1 2/ |
| | Protective flap | 1 2/ |
| | Neck tab | 2 |
| | Sleeve placket | 2 |



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| Material | Pattern parts | Cut parts |
|---|---|---|
| Plastic, foam, elastomeric (interlining) | Back upper - 1/8 inch | 1 |
| | Front and back yoke - 1/4 inch | 2 |
| | Back, upper trouser - 1/8 inch | 2 |
| | Back, lower trouser - 1/8 inch | 2 |
| | Center front trouser - 1/8 inch | 2 |
| | Side front, trouser - 1/8 inch | 2 |
| | Top sleeve - 1/8 inch | 2 |
| | Undersleeve - 1/8 inch | 2 |
| | Hood, side panel - 1/8 inch | 2 |
| | Hood, center panel - 1/8 inch | 1 |
| | Protective flap - 1/8 inch | 1 |
| Cloth, nylon, taffeta (lining) | Back, upper | 1 3/ |
| | Back trouser | 2 |
| | Front | 2 |
| | Top sleeve | 2 |
| | Undersleeve | 2 |
| | Hood, side panel | 2 |
| | Hood, center panel | 1 |
| Cloth, non-woven | Pocket welt interlining (bottom ply) | 2 |
| | Pocket welt interlining (top ply) | 2 |
| Chloroprene foam | Neck seal right | 1 |
| | Neck seal left | 1 4/ |
| | Neck seal center | 1 |

1/ Adjustment tabs may be stitched in strip form and cut to meet the finished dimensions of pattern parts.

*2/ Hood cord channel and protective flap shall be cut to accommodate specific size garment being cut.

3/ Upper back may be cut in two pieces with seam allowance added.

*4/ The tab end of the neck seal pattern part may be modified slightly (squared or rounded) to accommodate binding.

3.6 Construction.

3.6.1 Stitch, seam, and stitching types. Stitch, seam, and stitching types specified in Table I shall conform to FED-STD-751. Whenever two or more methods for seams or stitches are given for the same part of the operation, either one may be used. Where stitch type 401 is used, the looper or underthread shall be on the inside of the coverall. Seam allowances shall be maintained with seams properly sewn so that no raw edges, runoffs, twists, pleats, or open seams will result. Topstitching and edgestitching, unless otherwise specified, shall be 1/16 inch from the edge. Double needle stitching shall be 3/16 to 5/16 inch gauge unless otherwise specified.



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3.6.2 Thread breaks and ends of seams. The ends of seams and stitching produced with a 301 stitch type, when not caught in other seams and stitching, shall be backtacked not less than 3/8 inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks on 401 seaming may be repaired using 301 stitch type. Stitch type 401 shall not be used unless the ends are caught in other seams or stitchings.

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.7 Manufacturing operations requirements. The coveralls shall be manufactured in accordance with all operation requirements specified in Table I. The contractor is not required to follow the exact sequence of operations as specified.

* 3.8 Abbreviations in table of operations. The abbreviations used in Table I are as follows:

| | | |
|---|---|---|
| Stch | - | Stitch |
| in | - | inch |
| Ndl | - | Needle |
| Bob | - | Bobbin |
| Lpr | - | Looper |
| Mchne | - | Machine |
| Brtck | - | Bartack |
| Comrcl | - | Commercial |
| smlr | - | similar |
| Btnhl | - | Buttonhole |
| incl | - | including |
| dbl | - | double |
| chnstch | - | chainstitch |



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TABLE I - CONSTRUCTION OF COVERALLS

| NO. | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH IN | THREAD NDL BOB/ LPR |
|---|---|---|---|---|---|
| 1. | Cutting. | | | | |
| | *a. Cut the coveralls in strict accordance with patterns which show size, directional lines, and notches for the proper assembling of all parts. Neither the government patterns nor the working patterns shall be altered in any way, except that additional notches for use during construction are allowed on the working patterns. | | | | |
| | *NOTE: The protective flap and the hood cord channel shall be cut to accommodate the specific size garment being cut. | | | | |
| | *b. Except for the hood cord channel, the face side of the coated nylon shall be the uncoated side and shall be the exposed surface of the coverall. | | | | |
| | *c. Cut all component coated nylon parts out of one piece of material except adjustment tabs, facing, sleeve plackets, neck tabs, and inner and outer pockets which may be cut from ends. When so cut, the parts shall closely match the main assembly. Drill holes are prohibited. | | | | |
| | d. Directional lines on the fronts and back of outershell material may be swung from the warp direction by not more than 1-1/2 inches. Measurements shall be taken at the top and bottom edges of the pattern from the warp directional lines to the selvage edge (on the even side of the lay) of the fabric and the difference between the two measurements shall not exceed 1-1/2 inches. | | | | |



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|---|---|---|---|---|---|

e. All component parts of the uncoated nylon taffeta shall be cut from one piece of material, except the sleeves, which may be cut from ends. When cut from ends the shade of the sleeve linings shall be an approximate match.

*f. The component parts of the plastic or chloroprene foam need not match each other in shade and need not be cut from one piece of material.

2. Replacement of damaged parts.
Care shall be exercised during the spreading, cutting, and manufacturing operations to assure that material defects and damages specified in Section 4 are excluded and replaced with non-defective and properly matched material.

3. Size and shade marking.
a. All component parts of the coated fabric outershell and nylon lining shall be marked or ticketed to assure a uniform shade and size assembly throughout the coverall. Component parts which are allowed to be cut from ends need not be marked for shade, but shall be an approximate match to each other.

b. Any method of marking may be used except:
1. Metal fastening devices
2. Sew-on tickets
3. Adhesive type tickets whose adhesive causes discoloration or whose adhesive adheres to the material upon removal of the tickets.

c. The use of ink pad numbering machine, rubber stamp or pencil is allowed, provided the numbering is legible and does not show on the outside of the coverall.



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| NO. | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH IN | THREAD NDL | THREAD BOB/ LPR |
|---|---|---|---|---|---|---|
| 4. | Make interlining (plastic foam). | | | | | |
| | a. Cement all edges and butt all foam interlining to form the interlining shell, leaving sleeve seam open for slide fastener as indicated by pattern marks. Insure that all seams are properly cemented and are free of foreign matter prior to joining. | | | | | |
| | b. Stitch a continuous length of 1 inch wide bias-cut tape to the shell at the front opening overlapping 5/16 to 3/8 inch. Begin at the crotch and continue up and around the hood and down again to the crotch opening. The stitchline shall be 3/16 to 1/4 inch from the edge of the interlining shell. | 301 or 401 | LSa-1 | 6-9 | IB<br>IB | IB<br>IAA |
| | *c. Cement and tape (see 3.3.10.2) all butted foam interlining seams. | | | | | |
| | NOTE: Splices in the tape cover shall be permitted in long seams providing the splices are overlapped 3/4 inch and are no closer than two feet from each other. | | | | | |
| 5. | Make lining (nylon taffeta). | | | | | |
| | a. Stitch dart in each hood side panel, as indicated by marks on pattern, tapering and backtacking each pointed end of dart. | 301 | OSf-1 | 8-12 | IB | IB |
| | b. (1) Stitch, face to face, hood center panel to hood side panels. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge seams. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |
| | c. Stitch four (4) darts in trouser portion of fronts and four (4) darts in trouser portion of backs, as indicated by marks on pattern, tapering, and backtacking each pointed end of darts. | 301 | OSF-1 | 8-12 | IB | IB |



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| NO. | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH IN | THREAD NDL | THREAD BOB/ LPR |
|---|---|---|---|---|---|---|
| | d. (1) Superimpose coated side of front facings on face side of front linings. With front edges of facings and linings even, stitch all outer edges of facings 1/8 to 3/16 inch from raw edges. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge front edge of facings and linings. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |
| | e. (1) Stitch trouser backs, face to face, at seat. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge seam. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |
| | f. (1) For two-piece upper back, stitch center back seam, face to face. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge seam. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |
| | g. (1) Join upper back to trouser backs at the waist, face to face, forming a pleat at the center of the upper back. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge seam. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |
| | h. (1) Join the back to the fronts, face to face, at the shoulders. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge seams. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |



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| NO. | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH IN | THREAD NDL | THREAD BOB/ LPR |
|---|---|---|---|---|---|---|
| | *i. (1) Join topsleeve and undersleeve, face to face, at sleeve outseam leaving seam open for slide fastener as indicated by pattern marks. As an alternate, a strip of the basic coated cloth approximately 2-3/8 inches wide by 9 inches long may be used as a facing for the opening. The coated cloth side of the strip should be placed on the face side of the lining about 1-1/8 inches in from the cut edge to permit 3/4 inch of coated cloth extending for turning (oper. 6b). One row of stitching shall be placed approximately 1/8 inch from the back cut edge of the facing for holding purposes. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge seam. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |
| | j. (1) Join sleeves to body, face to face, at armholes. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge seams. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |
| | k. (1) Stitch sleeve inseam and body side seam, face to face, in one continuous operation. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge seams. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |
| | l. (1) Join fronts at crotch, face to face. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge seams. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |
| | m. (1) Stitch leg inseams, face to face, in one continuous operation. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge seam. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |



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TABLE I - CONSTRUCTION OF COVERALLS

| NO. | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH IN | THREAD NDL | THREAD BOB/ LPR |
|---|---|---|---|---|---|---|
| | n. (1) Join the hood to the body at neck, face to face, forming a pleat in the center of the upper back at neckline. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | (2) Overedge seam. | 503 or 504 or 602 | SSa-1 | 8-12 | IB | IB |
| | NOTE: Operations 5b, e, f, g, h, i, j, k, l, m, n, may be performed in one operation by simultaneously stitching and overedging. | 515 or 516 or 519 | SSa-2 | 8-12 | IB<br>IB<br>IB | IAA<br>IAA<br>IB |
| 6. | Join foam interlining to taffeta lining. | | | | | |
| | a. Join the foam interlining to taffeta lining at bottom of front closure (crotch), front edge of closure and hood facing in one continuous operation. Position the cotton tape (already attached foam interlining) on the front edge of prestitched coated fabric facing and taffeta lining. The raw edge of cotton tape shall extend 1/4 to 5/16 inch beyond the fold line on facing, as indicated by notches on pattern. Stitch tape, facing, and lining together 1/8 to 3/16 inch from the raw edge of tape. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | NOTE: When tape is sewn to hood opening, the raw edge shall be even, smooth, and flat. | | | | | |
| | *b. Turn foam interlining and taffeta lining to finished position. The legs and sleeves of taffeta lining should be inserted into the legs and sleeves of foam interlining. Stitch lining and interlining together at bottom of legs and sleeves approximately 1/4 inch from raw edge. Turn facing over sleeve openings to encase the foam at the sleeve openings and stitch approximately 3/8 inch from folded edge. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |



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|---|---|---|---|---|---|---|
| *7. | Make sleeve plackets. | | | | | |
| | a. Fold sleeve plackets face to face lengthwise and stitch 1/4 inch from folded side and bottom edges. Turn placket to finished position and top stitch 1/16 inch from folded edges. | 301 | SSe-2 | 8-12 (a&b) | IB | IB |
| *8. | Attach slide fastener to placket/sleeve. | | | | | |
| | a. Position the right side of the slide fastener tape even with the raw edges of the sleeve placket and stitch 1/8 inch from the edge through all plies. Top stop of slide fastener shall finish 1/2 inch from the bottom finished edge of sleeve placket. | 301 | SSa-1 | 8-12 | IB | IB |
| | b. Turn in seam allowance of lining even with the foam edges at sleeve openings. Lap the slide fastner tapes 1/4 inch over the foam edges and stitch 1/4 inch from back edges of slide fastener tape and 1/8 inch below the top stop. If alternate facing is used, place slide fastener chain 1/8 inch from folded edge and stitch 1/4 inch all around opening. | 301 | LSaj-1 | 8-12 | IB | IB |
| | NOTE: The sleeve placket shall finish on front of the sleeve and the bottom edge of the placket shall be even with the finished sleeve bottom. | | | | | |
| 9. | Make accessory/utility pocket. | | | | | |
| | a. Hem tops of the long and short utility pockets in accordance with notches, stitching 3/16 to 5/16 inch from folded edge. | 301 or 401 | EFa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | b. Place short utility pocket over long utility pocket with the bottom edges even and join with a single row of stitching down the center and length of short pocket thereby providing two openings for the short utility pocket. | 301 or 401 | SSv-1 | 8-12 | IB<br>IB | IB<br>IAA |



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|---|---|---|---|---|---|---|
| | c. Fold under the bottom and sides of the utility pocket 3/16 to 1/4 inch and stitch to the accessory pocket as indicated by marks on pattern, approximately 1/16 inch from folded edge. Reinforce with a second row of stitching parallel to the first row and approximately 1/8 inch from folded edge. | 301 or 401 | LSd-2 | 8-12 | IB<br>IB | IB<br>IAA |
| | d. Bartack the tops of all utility compartment openings, superimposed on the inner row of double stitching at corners and on the center row of single stitching. Bartacks shall be 1/4 to 3/8 inch long. | Brtck | | 10-16 per brtck | IB | IB |
| | e. Form a 3/4 inch bellows by joining the two bottom corners of the wide accessory pocket with a single row of stitching 1/4 inch from the raw edge. | 301 | SSa-1 | 8-12 | IB | IB |
| | f. Stitch, face to face, wide accessory pocket to narrow accessory pocket at top and bottom 7/16 to 9/16 inch from raw edges leaving an opening for the slide fastener. Fastener shall be centered on length of seamline. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | g. Join accessory pocket and slide fastener together by stitching around fastener opening 1/4 to 5/16 inch from folded side edges and from top and bottom stops of fastener. The slide fastener pull shall finish at the top of pocket when fastener is closed. | 301 or 401 | LSd-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | h. Die cut two 1/4 inch drainage holes as indicated by marks on pattern. | | | | | |
| 10. | Make sleeve and leg adjustment tabs. | | | | | |
| | a. Cut a 2-1/8 to 2-3/8 inch length of 2 inch wide hook fastener tape for the sleeve adjustment tab. | | | | | |



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|---|---|---|---|---|---|---|
| | *b. Cut a 3-3/8 to 3-5/8 inch length of 2 inch wide hook fastener tape for the leg adjustment tab. | | | | | |
| | c. Fold, back to back, and abut lengthwise edges of sleeve or leg adjustment tab (as applicable) at center of tab in accordance with fold lines on pattern. Fold lengthwise edges of tab in half again inwardly, face to face, forming a 2-3/16 inch wide four ply tab. Stitch the two lengthwise sides of tab 1/16 to 1/8 inch from folded edges. | 301 | Smlr to EFp-2 | 8-12 | IB<br>IB | IB<br>IAA |
| | d. Turn one end of tab under approximately 3/8 inch and place one end of the hook fastener tape over the raw edge and even with folded edge of tab end. Stitch all four sides of tape to tab 1/16 to 1/8 inch from edges of tape. | 301 | EFa and LSbj-1 | 8-12 | IB | IB |
| 11. | Make waistband adjustment tabs.<br>a. Cut two strips of basic material approximately 3-3/4 inches wide by 12-3/4 inches long. | | | | | |
| | *b. Cut one strip of 3/4 inch wide elastic webbing 6 inches long for sizes small and medium, 8 inches for sizes large and X-large, and 10 inches for size XX-large. | | | | | |
| | c. Fold the 12-3/4 inch long strips for waistband tabs in half lengthwise, back to back, and turn under the raw edges as indicated by marks on pattern. Stitch folded edges 1/16 to 3/16 inch from edge. The strip shall finish 1-1/16 to 1-3/16 inches wide. The ends shall not be stitched closed. | 301 | EFp-2 | 8-12 | IB | IB |
| | d. Insert one end of the elastic webbing approximately 1 inch into one end of each waistband length. Secure by stitching with a 5/8 inch cross boxstitch. | 301 | Cross-boxstitch | 8-12 | IB | IB |



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|---|---|---|---|---|---|---|
| | e. Cut two 3-3/8 to 3-5/8 inch lengths of 1 inch wide hook fastener tape. | | | | | |
| | f. Fold under the raw end of each waistband adjustment tab approximately 3/8 inch. Position the hook fastener tape over the folded under raw edge and stitch all sides of the tape 1/16 to 1/8 inch from the edges. | 301 | LSbj-1 | 8-12 | IB | IB |
| 12. | Make hood (coated fabric outershell). | | | | | |
| | a. Stitch darts in hood side panels, as indicated by marks on pattern, tapering and backtacking each pointed end of the darts. | 301 | OSf-1 | 8-12 | IB | IB |
| | b. Stitch, face to face, hood center panel to hood side panel 1/4 to 3/8 inch from raw edge. | 301 or 401 | LSq-2(a) | 8-12 | IB | IB |
| | c. Turn and topstitch 1/16 to 1/8 inch from fold with topstitch on center panel. | 301 or 401 | LSq-2(b) | 8-12 | IB<br>IB | IB<br>IAA |
| | d. Turn under ends of hood cord channel, back to back, 1/2 to 3/4 inch after adjusting to proper length to accommodate the size being manufactured and stitch 3/16 to 5/16 inch from folded edge. | 301 or 401 | EFa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | e. Cut a 44 to 46 inch length of drawcord. | | | | | |
| | f. Fold the hood cord channel, back to back, lengthwise with the length of cord inserted between. Position the raw edges of the hood cord channel on the face side of the hood face opening, as indicated on the patterns, and stitch 3/16 to 5/16 inch from raw edge. | 301 or 401 | SSav-2(a) with cord | 8-12 | IB<br>IB | IB<br>IAA |
| | g. Knot each end of cord. | | | | | |



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|---|---|---|---|---|---|---|
| 13. | Make bi-swing upper back. | | | | | |
| | a. Fold the back panel along the outer pattern marks, face to face, from the shoulder down to the bottom edge of back. Stitch with a single row of stitching approximately 1/16 inch from the folded edge. | 301 or 401 | OSf-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | b. Fold the back panel along the inner pattern marks, with the face side out, and stitch with a double row of 1/4 inch stitching. The outer row of stitching shall be approximately 1/16 from the folded edge. | 301 or 401 | OSf-2 | 8-12 | IB<br>IB | IB<br>IAA |
| | c. To complete bi-swing back, stay stitch across the pleat at top and bottom edges. | | | | | |
| 14. | Prepare front pockets and pocket welts. | | | | | |
| | a. Fold pocket welt pieces lengthwise, back to back, as indicated by marks on pattern, with the non-woven interlining inserted between the plies and stitch the back raw edge along length of welt. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | b. Die cut 1/4 inch diameter drainage holes in inner and outer pocket plies as indicated by pattern marks. | | | | | |
| | c. Stitch together the inner and outer pocket pieces, face to face, at the sides and bottom, 1/4 to 1/2 inch from the raw edge. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| 15. | Join pocket and welt to front. | | | | | |
| | a. Place bottom welt piece on face side of front on front edge of pocket opening with the inner pocket piece on top of welt as indicated by pattern marks. Stitch 1/2 to 5/8 inch from the raw edge. | 301 or 401 | Similar to SSbf-3(b) | 8-12 | IB<br>IB | IB<br>IAA |



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|---|---|---|---|---|---|---|
| | b. Cut the front, as indicated by pattern marks, tongue notching each end. Turn pocket and bottom welt piece through the opening and topstitch 1/16 to 1/8 inch from folded edge, taking care not to stitch through inner pocket piece. | 301 | Similar to SSbf-3(c) | 8-12 | IB | IB |
| | c. Turn under the raw edges of the pocket opening (tongues and facing) so that the opening is 3/4 to 7/8 inch wide. Place top welt into finished position and stitch the 3 sides (top, back, and bottom) 1/16 to 1/8 inch from folded edge of front catching front, welt, and pocketing.<br><br>NOTE: The welts shall finish 3/4 to 7/8 inch wide. The top welt shall be positioned so that the pocket opening faces the front of the coverall. | 301 | Similar to LSaj-1 | 8-12 | IB | IB |
| 16. | Prepare fronts and trouser backs.<br>a. Stitch darts in legs of front and in legs and waist of backs as indicated by marks on pattern, tapering and backtacking each pointed end of the darts. | 301 | OSf-1 | 8-12 | IB | IB |
| | b. Die cut 1/4 inch diameter holes in trouser legs as indicated by pattern marks. | | | | | |
| | c. Join leg adjustment tab to front as indicated by marks on pattern, stitching 3/16 to 5/16 inch from unfinished end of tab. | 301 or 401 | LSbl-2(a) | 8-12 | IB<br>IB | IB<br>IAA |
| | d. Fold tab back over stitchline and secure back end of tab to leg with a 1 to 1-1/2 inches long by full width of tab boxstitch. The top and bottom rows of stitching shall be superimposed on existing lines of stitching. Back row of stitching shall be 1/16 to 1/8 inch from back folded edge. | 301 | Boxstitch | 8-12 | IB | IB |



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|---|---|---|---|---|---|---|
| | e. Stitch backs of trousers together, face to face, at seat seam. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | f. Cut two 7-1/2 to 7-3/4 inch long, 1 inch wide loop fastener tapes. Join to fronts as indicated by pattern marks, stitching all four sides, 1/16 inch from the edge, catching tops of hanging pockets. | 301 or 401 | LSbj-1 | 8-12 | IB<br>IB | IB<br>IAA |
| 17. | Join upper back to trouser backs.<br>Stitch the upper back to the trouser backs at the waist, face to face. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| 18. | Join waistband adjustment tab and back waist channel to back.<br>a. Tack or bartack, 5/8 to 3/4 inch long, elastic webbing to the back with the waistband adjustment tab centered on the center seat seam and the waist seam (fastener tape side down). | 301 or Bar-tack | | 8-12<br>21-28 per bartack | IB<br>IB | IB<br>IB |
| | b. Turn under the ends of the back waist channel approximately 5/8 inch and topstitch 3/8 to 1/2 inch from the folded edges. | 301<br>401 | EFa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | c. Turn under the raw side edges of back waist channel 3/8 inch and double stitch each folded edge to the back. The back waist channel shall be centered on the seat and waist seams and shall finish 2-3/8 to 2-5/8 inches wide. | 301 or 401 | SSat-4 | 8-12 | IB<br>IB | IB<br>IAA |
| 19. | Join shoulders.<br>Join the back to the fronts, face to face, at the shoulders. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |



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|---|---|---|---|---|---|---|
| 20. | Join accessory/utility pocket to the left sleeve. | | | | | |
| | Turn under all raw edges of the accessory/utility pocket 1/4 to 5/16 inch. Position on left topsleeve, as indicated by marks on pattern and join to sleeve with a double row of 1/16 to 1/8 inch gauge stitching. | 301 or 401 | LSd-2 | 8-12 | IB<br>IB | IB<br>IAA |
| | NOTE: The pocket shall finish 4-5/8 to 4-7/8 inches wide by 7-3/4 to 8-1/4 inches long. | | | | | |
| 21. | Join top and undersleeves and attach sleeve adjustment tapes. | | | | | |
| | *a. Join topsleeve and undersleeve, face to face, at sleeve outseam leaving sleeve outseam open for slide fastener as indicted by pattern marks. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | b. Join adjustment tab to topsleeve, as indicated by marks on pattern, by stitching 3/16 to 5/16 inch from unfinished end of tab. Finished end of tab shall point towards back of coverall. | 301 or 401 | LSbl-2(a) | 8-12 | IB<br>IB | IB<br>IAA |
| | c. Fold adjustment tab back over stitchline and secure back end of tab to sleeve with a 1 to 1-1/2 inch long by full width of tab boxstitch. The top and bottom rows of stitching shall be superimposed on existing lines of stitching. Back row of stitching shall be 1/16 to 1/8 inch from folded edge. | 301 | Boxstitch | 8-12 | IB | IB |
| | d. Stitch a 5 to 5-3/4 inch long by 2 inch wide loop fastener tape to the sleeve, as indicated by marks on pattern, stitching all four sides 1/16 inch from edge. | 301 | LSbj-1 | 8-12 | IB | IB |
| | e. Die cut 1/4 inch diameter drainage holes at bottom of sleeve as indicated by marks on pattern. | | | | | |



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|---|---|---|---|---|---|---|
| 22. | Join sleeves to body. | | | | | |
| | Join sleeves to body, face to face, at armholes. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| 23. | Join sleeve inseams, body, and leg outseams. | | | | | |
| | a. Stitch sleeve inseams, body, and leg outseams, face to face in one continuous operation. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | b. Stitch a 9-1/4 to 9-3/4 inches long by 2 inches wide loop fastener tape to the legs, as indicated by marks on pattern. All four sides of tape shall be stitched 1/16 inch from the edges. | 301 | LSbj-1 | 8-12 | IB | IB |
| 24. | Join crotch seam. | | | | | |
| | Join fronts, face to face, at the crotch seam. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| 25. | Join leg inseams. | | | | | |
| | Join leg inseams, face to face, in one continuous operation. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| 26. | Join hood to body. | | | | | |
| | a. Join hood to the body, face to face, at neckline. | 301 or 401 | SSa-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | *b. Center a 1 inch wide by 2 inch long piece of hook fastener tape on the neckline seam of the right front with the outside end of the tape 1/2 inch from the finished front edge and the long side parallel to the neckline seam. Stitch sides of the tape 1/16 to 1/8 inch from edge. | 301 | LSbj-1 | 8-12 | B | B |
| 27. | Join combined foam interlining and taffeta lining to outer shell. | | | | | |
| | * NOTE: The foam interlining and taffeta lining shall be combined to make the inner shell. | | | | | |
| | a. Join the hood inner shell to the hood outer shell face to face. | 301 or 401 | SSax-2(a) (with cord) | 8-12 | IB | IB |



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|---|---|---|---|---|---|---|
| | b. Turn to finished position and topstitch, 3/16 to 1/4 inch from the seamline. | 301 or 401 | SSax-2(b) (with cord) | 8-12 | IB | IB |
| | *c. Place body inner shell and body outer shell into finished position (inner shell legs and arms into outer shell legs and arms). Notch outer shell seam allowance for sleeve opening as indicated by pattern marks and turn in seam allowance along both edges. Hem the sleeve and leg bottoms as follows: | | | | | |
| | (1) Turn under the raw edge of outer shell 1-1/4 inches.<br>(2) Place the bottom edges of inner shell between the plies of outer shell up to the fold line. | | | | | |
| | (3) Turn under the raw edge of outer shell approximately 1/4 inch and stitch 1/16 to 1/8 inch from folded edge through all plies. | 301 or 401 | SS1-1 | 8-12 | IB<br>IB | IB<br>IAA |
| | *d. Turn under the self facings of the inner shell and the outer shell, as indicated by pattern marks. Position the slide fastener between the plies with all folded fabric and fastener edges aligned. Topstitch 1/2 to 3/4 inch from the folded edges. Beginning at the top of the slide fastener down the side, and around the crotch, finishing at the top of the slide fastener on the other side.<br><br>NOTE: The top stop of the slide fastener shall finish not more than 1/4 inch from the finished neck seam. | 301 | LSe-1 | 8-12 | IB | IB |
| | *e. Topstitch 3/8 to 1/2 inch from folded edge of sleeve openings, along both sides and across the bottom. Stitching shall go thru all plies and secure the slide fastener tape and the top end of sleeve placket. | 301 | LSaj-1 | 8-12 | IB | IB |



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|---|---|---|---|---|---|---|
| 28. | Make and attach the front closure protective flap. | | | | | |
| | a. Fold the protective flap in half lengthwise, face to face, and stitch the top edge only. | 301 | Sa-1 | 8-12 | IB | IB |
| | b. Turn flap to finished position and place the foam interlining piece between the plies and stitch across the top folded edge down the full length of the back raw edge without catching the foam interlining and across the bottom raw edge, 1/8 to 3/16 inch from the edge. | 301 | SSa-1 | 8-12 | IB | IB |
| | c. Stitch the protective flap to the inside of the right front and hood, 1/16 to 1/8 inch from the raw edge, with the line of stitching superimposed on the stitching which secures the slide fastener to the right front. The top edge of the protective flap shall finish 1/8 to 3/8 inch above the open end of the hood cord channel. | 301 | Similar to LSa-1 | 8-12 | IB | IB |
| | d. Secure the bottom portion of the protective flap in the crotch area with a 5/8 to 3/4 inch horizontal bartack with the stitching superimposed on the stitching which secures the bottom of slide fastener. When necessary, trim excess from bottom of protective flap. | Bar-tack | | 21-28 per bartack | IB | IB |
| *29. | Make neck tab. | | | | | |
| | a. Position the neck tab pieces back to back with the loop fastener tape positioned and shaped according to pattern marks on the bottom ply. Double stitch through all plies, with a 1/8 to 1/4 inch stitch gauge and the outside row of stitching 1/8 inch from the raw edges. | 301 | SSa-2 | 8-12 | IB | IB |



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| *30. | Attach neck tab and fastener tape. Positon neck tab on the left front according to pattern marks and in the direction to mate with the hook fastener tape on the right side of the neckline. Turn under raw edge of tab 1/4 inch and stitch to front. Folded edge of neck tab shall finish approximately 1/2 inch from finished front edge of the coverall. | 301 | LSbk-2 | 8-12 | IB | IB |
| *31. | Assemble neck seal. | | | | | |
| | a. Cement (abut) the right and left neck seal pieces to the center neck seal piece. Insure that all seams to be cemented are free of foreign matter prior to joining. The skin side of the foam shall be considered the face. Bind edges with 1-7/8 inch bias binding tape. | 301 | BSe-1 | 8-12 | IB or IB | IAA |
| | b. At center back, superimpose top edge of identification label on binding at bottom edge of neck seal and stitch across top edge of label. If not backtacked, stitching shall extend a minimum of three stitches. | 301 | LSbj-1 | 8-12 | IB | IB |
| | c. Position a 4-1/2 inch long and a contour piece (according to pattern marks) of 2 inch wide hook fastener tape to the face side of the right neck seal in accordance with the marks on patterns. Stitch all sides of the tape 1/16 to 1/8 inch from the edge. | 301 | LSbj-1 | 8-12 | IB | IB |
| | d. Cut a 1-1/2 inch long piece of 2 inch wide loop fastener tape shaped to fit the point of the left neck seal. Position that piece and a 4-1/2 inch long piece of 2 inch wide loop fastener tape to the back side of the left neck seal in accordance with pattern marks. Stitch all sides of tape 1/16 to 1/8 inch from edge. | 301 | LSbj-1 | 8-12 | IB | IB |



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TABLE I - CONSTRUCTION OF COVERALLS

| NO. | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH IN | THREAD NDL | THREAD BOB/ LPR |
|---|---|---|---|---|---|---|
| *32. | Attach neck seal to the lining. | | | | | |
| | a. Position neck seal on the neckline of the lining assembly face to face matching marks on the top edge of the center neck seal with the lining neckline seam and cemented seams of neck seal with shoulder seams. Stitch seal to neckline in accordance with pattern marks to the lining only. | 301 | Similar to SSbd-1 | 8-12 | IB | IB |
| | b. Line up inside edges of right and left neck seals with the inside edge of the facings and edge stitch to the facing and lining only. | 301 | LSbz-3(b) | 8-12 | IB | IB |
| | c. Stitch the right and left neck seals through all plies from the neckline to the bottom edge of the neck seal. Stitching shall be approximately 2-1/2 inches from the finished front edge of the coverall. | 301 | SSa-1 | 8-12 | IB | IB |
| | d. Position a 1 inch wide by 2 inch long piece of loop fastener tape to the inside of the right facing with the long outside edge parallel to and not more than 1/2 inch from the neck seal seam. The top end of the tape shall be approximately 2 inches from the finished neckline seam. Stitch all sides of the tape 1/16 to 1/8 inch from the edge thru all plies. | 301 | LSbj-1 | 8-12 | IB | IB |
| | e. Position a 1 inch wide by 2 inch long piece of hook fastener tape to the inside of the left facing with the long outside edge parallel to and not more than 1/2 inch from the neck seal seam. The top end of the tape shall be approximately 2 inches from the finished neckline seam. Stitch all sides of the tape 1/16 to 1/8 inch from the edge thru all plies. | 301 | LSbj-1 | 8-12 | IB | IB |



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TABLE I - CONSTRUCTION OF COVERALLS

| NO. | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH IN | THREAD NDL BOB/ LPR |
|---|---|---|---|---|---|
| 33. | Clean.<br>a. Trim all ends of stitching and remove loose threads.<br><br>b. Remove all spots, stains, shade, and size tickets.<br><br>*c. All slide fasteners and hook and loop fastener tabs shall be in a closed postion. The left neck seal shall be folded onto itself to attach to the hook fastener tape on the left front facing. The right neck seal shall attach to the loop fastener tape on the right front facing. | | | | |



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* 3.8 Measurements. Measurements of finished coveralls shall be as shown in Table II. All measurements and tolerances are expressed in inches.

Table II - Sizes and finished measurements

| Size | Chest A | Leg Inseam B | Length C | Sleeve Inseam D |
|---|---|---|---|---|
| Small (34-36) | 47 | 27 | 59 | 19-1/2 |
| Medium (38-40) | 51 | 28 | 61 | 20-1/2 |
| Large (42-44) | 55 | 29 | 63 | 21-1/2 |
| X-Large (46-48) | 59 | 30 | 65 | 22-1/2 |
| XX-Large (50-52) | 63 | 31 | 67 | 23-1/2 |
| Tolerance | ± 1 | ± 3/4 | ± 1 | ± 3/4 |

NOTE: Refer to Figure 1 for A, B, C, and D measurements. The coverall shall be laid flat with the front slide fastener completely closed.

A. Measure across front at base of armholes from folded edge to folded edge. Double the value obtained.
B. Measure from crotch seam along inseam to bottom of hemmed leg.
C. Measure from intersection of shoulder and hood seams to bottom of hemmed leg.
D. Measure from base of armhole along inseam to bottom of sleeve.

3.9 Workmanship. The finished coveralls shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.



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4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 6.2), the first article submitted in accordance with 3.2 shall be visually inspected as specified in 4.4.2 for compliance with design, construction, workmanship, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

* 4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. A certificate of compliance shall be furnished for the foam requirements of 3.3.13.1, knit fabric requirements of 3.3.13.2, the colorfastness to laundering requirements of 3.3.13.3, 3.3.5.4, and 3.3.5.5 and the colorfastness to crocking requirements of 3.3.7 when tested in accordance with 4.4.1.3. In addition, the expanded plastic foam shall be tested for buoyancy as specified in 4.4.1.1 and MIL-P-12420, except that the sample unit submitted for testing shall be 10 square feet Three determinations shall be performed per sample unit of foam and the results reported to the nearest 0.1 pound. Component listed in Table III shall be tested for characteristics shown therein. The methods of testing as specified herein shall be followed. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The basis for lot size for laminated chloroprene foam shall be one sheet. The sample unit for testing shall be one square yard.

The sample size shall be in accordance with the following:

| Lot size | Sample size |
|---|---|
| 800 or less | 2 |
| 801 thru 22,000 | 3 |
| 22,001 and over | 5 |



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Table III - Testing of components

| Components | Characteristic | Determi-nations | Requirement paragraph | Test method |
|---|---|---|---|---|
| Laminated foam | Adhesive strike thru | 1 | 3.3.13.4 | Visual |
| | Wrinkled or puckered | 1 | 3.3.13.4 | Visual |
| | Laundering resistance | 5 | 3.3.13.4 | 5556 1/ |
| | Adhesion: | | | |
| | Initial | 5 | 3.3.13.4 | 5950 2/ |
| | After 3 launderings | 5 | 3.3.13.4 | 5556 & 5950 2/ |
| | After aging | 5 | 3.3.13.4 | 5950 2/ 4/ |
| | Blocking | 3 | 3.3.13.4 | 5872 3/ |

1/ Laundering. Three cotton laundering cycles shall be performed on 8 by 8 inch specimens.

2/ Adhesions. Except that the distance between the clamps shall be two inches at the start of the test. Slitting the specimen between the nylon knit and foam to help separate the fabric from the foam sufficiently to fit the clamps is permissable. The adhesion shall be tested in the fabric's length (wale) direction. If the adhesion value is not measureable because the foam is tearing while the bond between the foam and the knit cloth remains intact, the bond shall be considered acceptable.

3/ Blocking. Except specimens 4 by 4 inches shall be placed together fabric side to fabric side.

4/ After aging. The specimen shall be suspended vertically in an air oven at 158° ± 2°F (70° ± 1°C) for seven days. At the conclusion of the aging interval the specimens shall be cooled at room temperature on a flat surface and allowed to rest for four hours before evaluating for adhesion.

4.4.1.1 Buoyancy. The buoyancy test shall be made with a sample of the sheet material measuring 12 by 12 inches by thickness of material furnished. Securely attach a spring scale, accurate to 0.25 pound, in a position directly over the test tank. Suspend a weighted wire basket from the scale in such a manner that the basket can be weighed while completely submerged in water. Proceed as follows:

a. Weigh the empty basket under water.

b. Place test sample inside the basket and submerge it so that the top of the basket is at least 2 inches below the surface of water. Allow sample to remain submerged for 24 hours.

c. After 24 hour submergence period, weigh the wire basket with the sample inside while both are still under water.

d. Compute buoyancy in pounds per cubic foot:

$$\text{Lbs/CuFt} = \frac{\text{Weight of empty basket in water - Weight of basket and sample in water}}{\text{Volume of foam material in cubic feet}}$$



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* 4.4.1.2 Seam strength. The cement shall be tested for seam strength properties in accordance with method 5100 of FED-STD-191 and the following exceptions: Cemented abutted seams shall be prepared using 4 by 6 inch test specimens. Abutted seams shall be cemented across the 4 inch dimension. The fully cured cemented seams shall be centered between and parallel to 2 inch long clamps. The clamps shall be 1 inch apart at start of test. The cement shall be unacceptable if one or more test specimens fail to meet the seam strength requirement specified in 3.3.10. If the adhesion value is not measurable because the foam itself is tearing while the seam remains intact, the seam strength shall be considered acceptable. A certificate of compliance shall be submitted for the identification requirement.

4.4.1.3 Colorfastness of tape, thong, and drawcord. The slide fastener tape and nylon thong shall show good colorfastness to laundering when tested in accordance with Test Method 5614 of FED-STD-191. The dyed polyester drawcord shall show colorfastness to crocking not numerically below Munsell Value 8.5 when tested in accordance with Test Method 5651 of FED-STD-191.

* 4.4.1.4 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are carried out as specified. The government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection had indicated non-conformance.

4.4.1.5 Intermediate inspection.

* 4.4.1.5.1 Visual examination. Defects found during the intermediate examination shall be classified in accordance with 4.4.1.5.1.1. The applicable inspection levels and acceptable quality levels shall be as indicated in 4.4.1.5.1.3.

* 4.4.1.5.1.1 Examination of foam interliner. The foam interliner assembly shall be examined for defects in cutting, fitting, and other construction characteristics which cannot be seen in the end item. The sample unit shall be one completely fabricated foam interliner assembly. The lot size shall be expressed in the same terms as the sample.

| Defect | Classification | | |
|---|---|---|---|
| | Critical | Major | Minor |
| Any hole, cut or tear in the foam | X | | |
| Construction not as specified | | X | |
| Any component missing or other than the type specified | | X | |
| Any component misplaced or not affixed as specified | | X | |
| Any open seam | X | | |
| Any seam not taped when specified | X | | |
| (Butted seam) twisted, puckered, or pleated | | | X |
| Thickness not as specified | | X | |
| Splicing of seam tape not as specified | X | | |



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* 4.4.1.5.1.2 Break strength of cemented seam. For each 100 interliners or less submitted for inspection, one completed fabricated foam interliner shall be selected for seam test. The seam test shall be as described in paragraph 4.4.1.2 and the sample seams taken from the following areas: knee seam, outside arm seam, hood side seam, leg outseam, and leg inseam. Any test failure shall be cause for rejection of the lot (see 3.3.10.1).

* 4.4.1.5.1.3 Foam interliner inspection levels and Acceptable Quality Levels. The inspection levels and acceptable quality levels (AQL's) expressed as defect per 100 units for visual examination shall be as follows:

| | AQL | Inspection level |
|---|---|---|
| Major | 2.5 | II |
| Total (major and minor combined) | 6.5 | II |

NOTE: If one or more critical defects are found during sampling inspection, the lot shall be rejected. For each lot found to be acceptable by the sampling plan, the balance of the lot shall be inspected for critical defects only. Any critical defect found shall be removed from the lot and replaced with a conforming item.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with provisions for finished coveralls as specified in MIL-STD-1668 and the following list:

| Defect | Point value |
|---|---|
| MATERIAL DEFECT(S) AND DAMAGE(S) | |
| a. Any size hole, needle chew, cut, tear, smash, visible mend of burn: | |
| (1) On outside | 3 |
| (2) On inside | 1 |
| b. Misweave, missing yarn, or broken yarn; more than 1/2 inch in any direction and on outside only | 2 |
| CLEANNESS | |
| Any marking exposed or readily visible on outside | 1 |
| SHADED PART(S) | |
| a. On outside: | |
| 1. Variation in shade between two or more parts (except parts which may be cut from ends) | 1 |
| 2. Variation in shade within a part (except parts which may be cut from ends | 1 |
| 3. Part cut from ends badly shaded | 1 |



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| Defect | Point value |
|---|---|
| SEAM(S) AND STITCHING(S) | |
| a. Accuracy of seaming | |
| 1. Seam puckered, twisted, or pleated (unless otherwise classified herein) | 2 |
| *2. End of seam or stitching when not caught in another seam or stitching backtacked less than 3/8 inch except label stitching | 1 |
| 3. Any seam restitched damaging material to a degree where it may tear along the row of stitching | 3 |
| * b. Gauge of stiching | |
| 1. Top or edge stitching seam too close to edge resulting in damage to material: | |
| (a) Affecting appearance | 1 |
| (b) Affecting serviceabilty | 3 |
| c. Raw edges | |
| 1. On outside: | |
| (a) -up to 1/2 inch inclusive | 2 |
| (b) -more than 1/2 inch | 3 |
| 2. On inside - more than 1/2 inch | 1 |
| d. Open seam (see NOTE:) | |
| 1. On outside: | |
| (a) - up to 1/4 inch inclusive | 1 |
| (b) - more than 1/4 inch up to 1/2 inch inclusive | 2 |
| (c) - more than 1/2 inch | 3 |
| 2. On inside: | |
| (a) - up to 1/2 inch inclusive | 1 |
| (b) - more than 1/2 inch | 2 |
| NOTE: One or more broken stitches or two or more continuous skipped or runoff stitches constitute an open seam. | |
| e. Seam and stitch types | |
| 1. Looper thread on outside or ends not caught in other seams or stitchings (when 401 stitch is used) | 3 |



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| Defect | Point value |
|---|---|
| f. Stitch tension | |
| Loose tension resulting in loose seam | 3 |
| g. Bartack(s) | |
| 1. Missing, insecure, misplaced, or not serving intended purpose | 3 |
| 2. Not size specified | 1 |
| LABEL | |
| Missing, incorrect, or illegible | 3 |
| LINING | |
| a. Too full or too short | 3 |
| b. Twisted or pleated, affecting outside appearance | 1 |
| UTILITY/ACCESSORY POCKET | |
| a. Attached to right sleeve instead of left sleeve | 3 |
| b. Slide fastener opening not facing toward front of coveralls | 3 |
| c. Utility compartments not an integral part of combination pocket | 2 |
| d. One utility compartment missing | 1 |
| e. Two or more utility compartments missing | 3 |
| f. Accessory pocket not formed with bellows | 1 |
| FACING (INSIDE AND OUTSIDE) | |
| a. Twisted, puckered, causing excessive fullness | 1 |
| b. Facings do not lay flat when slide fastener is in closed position | 2 |
| PROTECTIVE FLAP | |
| a. Twisted, tight or full; not permitting flap to lie smooth | 3 |
| b. Top of flap less than 1/8 or more than 3/8 inch above hood cord channel opening | 1 |



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| Defect | Point value |
|---|---|
| HOOD | |
| a. Not topstitched as specified | 1 |
| b. Channel not attached as specified | 2 |
| * c. Ends of drawcord not treated with a clear resin | 2 |
| WAISTBAND AND BACK WAIST CHANNEL | |
| a. Elastic portion missing | 3 |
| b. Hook fastener tape not on inside of waistband | 3 |
| c. Channel not as specified | 1 |
| d. Channel not centered over waist seam | 1 |
| e. Finished width of waistband or back waist channel not as specified | 1 |
| WELT POCKETS | |
| a. Welt poorly formed or irregular in width | 1 |
| b. Welt twisted, puckered or distorted | 1 |
| c. Pocket opening does not face towards front of coverall | 2 |
| d. Welts finish less than 3/4 inch or more than 7/8 inch wide | 1 |
| FRONT OR SLEEVE SLIDE FASTENER | |
| a. Top stop: | |
| - on front set more than 1/4 inch from finished neck seam | 2 |
| - on sleeve set more than 1/2 inch from finished bottom edge of sleeve placket | 2 |
| b. Ends of slide fastener tape not securely caught in stitching | 3 |
| NOTE: Slide fastener shall be checked for proper function by opening and closing slide fastener at least three times. | |
| DRAINAGE HOLES | |
| a. Any missing on pockets, sleeve or leg bottoms | 2 |
| b. Not size specified | 1 |



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| Defect | Point value |
|---|---|
| c. Not specified distance from sleeve or leg bottom | 1 |
| NECK SEAL | |
| a. Seam not butted and cemented properly, i.e., not smooth and flat | 3 |
| b. Not stitched to coverall as specified | 3 |
| c. Hook and loop not positioned as specified, preventing a tight seal | 3 |
| d. Binding tape other than bias cut | 1 |
| e. Binding tape not turned under and positioned evenly on neck seal for more than 1 inch | 1 |

4.4.3 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units in accordance with MIL-STD-105.

| Examine | Defect |
|---|---|
| Marking (exterior and interior | Omitted, incorrect, illegible, of improper size, location, sequence, or method of application. |
| Materials | Any component missing, damaged, or not as specified. |
| Workmanship | Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container. |
| Content | Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container. 1/ |

1/ For this defect, one item from each shipping container in sample shall be examined.



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4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspected lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 in accordance with MIL-STD-105.

| Examine | Defect |
|---|---|
| Finished dimensions | Length, width,or height exceeds specified maximum requirements. |
| Palletization | Pallet pattern not as specified.<br>Interlocking of loads not as specified.<br>Load not bonded with required straps as specified. |
| Weight | Exceeds maximum load limits. |
| Marking | Omitted, incorrect, illegible, of improper size, location, sequence, or method of application. |

5. PACKAGING

5.1 Preservation-packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each pair of coveralls, with the slide fasteners closed, shall be laid flat, front down with sleeves and legs fully extended. Bring the legs up over the back approximately 25 inches. Fold the sleeves at right angles across the back. The length shall then be folded in half with the hood brought down over the chest. Each completely folded coverall shall measure approximately 23 by 15 inches. Each pair of folded coveralls shall be inserted in a close-fitting clear polyethylene film bag having a thickness of 0.003 inch (± 20 percent tolerance). The bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the heat seal made as close as possible to the open end. A 1/4 inch diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape. Alternately, the final closure of the polyethylene bag may be accomplished by means of a tuck or reverse flap.

5.1.2 Level C. Coveralls shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The package may be the same as that normally used by the supplier for retail distribution when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).



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p* 5.2.1 Level A. Three coveralls of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class weather resistant, variety DW, grade V15c, size 3A of MIL-B-17757. The fiberboard for the liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat, two in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

* 5.2.2 Level B. Three coveralls of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to type CF, class domestic, variety DW, grade 200, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, of MIL-B-17757. Level A packages shall be packed flat, two in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

5.2.3 Level C. Item packaged, as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Polyethylene bagged packages. Polyethylene bagged packages shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

* 5.4 Palletization. When specified (see 6.2) item packed as specified shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K, and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

The pallet shall be 4-way, Type IV; Type V, Class 1, Size 2; or Type VIII, fabricated from wood group I, II, III or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 fabricated from wood groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.



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6. NOTES

6.1 Intended use. The coveralls are intended to be worn by Navy personnel as anti-exposure suits to protect the wearer against any combination of wind, wet, and cold conditions.

6.2 Ordering date. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Sizes required (see 1.2)

*c. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, testing and approval of the first article.

d. Levels of packaging and packing required (see 5.1 and 5.2).

*e. When palletization is required (see 5.4).

6.3 Samples and patterns. For access to samples and patterns address the pprocuring office issuing the invitation for bids (see 3.1 and 3.5).

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 Subject term (key word) listing.

Clothing, anti-exposure
Coverall
Coverall, foam, chloroprene
Coverall, foam, interlining
Foam interlining
Twill, coated nylon

Custodian:
Navy - NU

Preparing Activity:
Navy - NU

Review Activity:
DLA - CT

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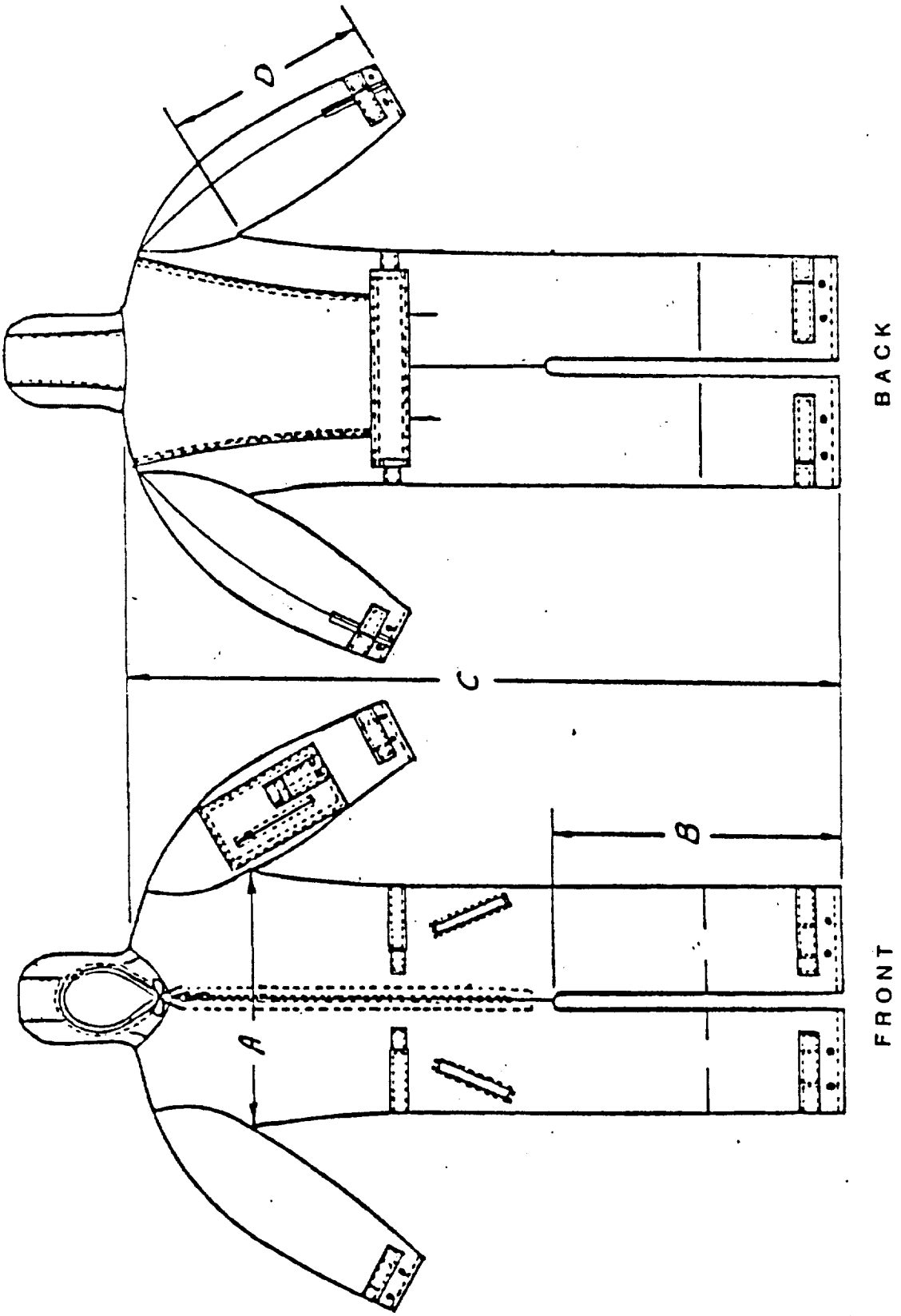



FIG 1 COVERALLS, ANTI-EXPOSURE



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-29109B | COVERALLS, ANTI-EXPOSURE |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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